

C Programming

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PREPROCESSOR STATEMENTS

The *define* statement is used to make programs more readable, and allow the inclusion of macros. Consider the following examples,

```
#define TRUE      1      /* Do not use a semi-colon , # must be first charact
#define FALSE     0
#define NULL      0
#define AND       &
#define OR        |
#define EQUALS    ==

        game_over = TRUE;
        while( list_pointer != NULL )
            .....
```

Macros

Macros are inline code which are substituted at compile time. The definition of a macro, which accepts an argument when referenced,

```
#define SQUARE(x)  (x) * (x)

y = SQUARE(v);
```

In this case, *v* is equated with *x* in the macro definition of *square*, so the variable *y* is assigned the square of *v*. The brackets in the macro definition of *square* are necessary for correct evaluation. The expansion of the macro becomes

```
y = (v) * (v);
```

Naturally, macro definitions can also contain other macro definitions,

```
#define IS_LOWERCASE(x)  (( (x) >= 'a') && ( (x) <= 'z') )
#define TO_UPPERCASE(x)  (IS_LOWERCASE (x) ? (x) - 'a' + 'A' : (x))

while(*string) {
    *string = TO_UPPERCASE(*string);
    ++string;
}
```

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